pi-labs.ai at ArA-DF 2026: Score-Level Fusion of Self-Supervised Speech Encoders with an AASIST Back-End for Arabic Audio Deepfake Detection

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Abstract

This paper presents our submission to the ArA-DF 2026 Shared Task on Arabic Audio Deepfake Detection, focusing on Track 1, which evaluates robustness to previously unseen Arabic dialects. Our proposed system combines two complementary self-supervised speech encoders, XLS-R-300M and WavLM-Large, each integrated with an AASIST graph-attention back-end for spoofing detection. The final prediction is obtained through score-level fusion of the two independently trained models, enabling the system to leverage the complementary representations learned by each encoder. To improve robustness to domain shift, we introduce two complementary adaptation strategies: telephony channel augmentation to reduce the mismatch between the training and evaluation domains, and an iterative pseudo-label self-training framework that exploits high-confidence predictions on unlabeled evaluation data for progressive model refinement. On the official ArA-DF 2026 Track 1 evaluation set, our system achieved an Equal Error Rate (EER) of 0.9265%, ranking 4th on the official leaderboard. These results demonstrate that combining complementary self-supervised speech representations with domain-aware adaptation techniques provides an effective and robust solution for cross-dialect Arabic audio deepfake detection.

1 Introduction

Recent advances in neural speech synthesis and voice conversion have made synthetic speech increasingly realistic, creating significant challenges for reliable audio deepfake detection. A key difficulty is domain generalization, where variations in recording conditions, synthesis methods, and language characteristics degrade performance.

The ArA-DF 2026 Shared Task addresses this problem through Track 1 (Dialect Generalization), requiring systems to detect spoofed speech from previously unseen Arabic dialects. The primary challenge lies in maintaining reliable detection performance under domain shifts caused by unseen dialects and telephony recording conditions. To address this challenge, we propose a score-level fusion framework based on complementary self-supervised speech encoders and an AASIST back-end (Jung et al., 2022). To improve robustness under domain shift, we further employ telephony channel augmentation and iterative pseudo-label self-training using high-confidence unlabeled evaluation data. Our results suggest that combining complementary SSL representations with domain adaptation techniques is effective for improving robustness to cross-dialect distribution shifts.

Our main contributions are:

- A score-level fusion framework combining complementary SSL encoders with an AASIST back-end for Arabic audio deepfake detection.
- Telephony-aware augmentation and iterative pseudo-label self-training for improved cross-dialect robustness.
- Achieved an Equal Error Rate (EER) of **0.9265%**, which is only 0.05 higher compared to the top-1 system, ranking **4th** on the official ArA-DF 2026 Track 1 leaderboard.

2 Background and Task Setup

2.1 Task

We participate in Track 1 (Dialect Generalization) of the ArA-DF 2026 Shared Task, which evaluates audio deepfake detection under previously unseen Arabic dialects. Given an input speech recording, the objective is to assign a spoofing score to distinguish bona fide from spoofed speech. The primary challenge is to maintain reliable detection performance when the dialect and recording conditions

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differ from those observed during training. Performance is evaluated using the Equal Error Rate (EER).

2.2 Data

Our system is trained on the ArA-DF training corpus (ArabicSpeech, 2026), which contains approximately 43.5K samples in the labelled train-development split. To prevent leakage between segments originating from the same recording, we reconstructed the development partition using recording-level separation. Following segmentation, the resulting model-development split comprised 86,627 training segments and 13,428 validation segments, including 5,799 bona fide and 7,629 spoofed validation samples.

2.3 Models and Related Work

Recent audio deepfake detection systems increasingly combine self-supervised representations with graph-based anti-spoofing back-ends. AASIST (Jung et al., 2022) has demonstrated strong performance by modeling spectro-temporal relationships using graph-attention, while self-supervised speech models (Baevski et al., 2020; Babu et al., 2022; Chen et al., 2022) have demonstrated strong transferability across spoofing detection tasks. Motivated by prior SSL+AASIST approaches (Tak et al., 2022), we extend this paradigm by combining two complementary SSL encoders through score-level fusion and incorporating domain-adaptation strategies tailored to cross-dialect Arabic deepfake detection.

3 System Overview

Figure 1 shows the proposed dual-branch framework, in which XLS-R-300M+AASIST and WavLM-Large+AASIST independently produce spoofing scores that are subsequently normalized and fused to obtain the final prediction.

3.1 Dual SSL Encoders with AASIST Back-Ends

Although both branches employ Transformer-based SSL encoders, they differ in their pretraining objectives. XLS-R is pretrained using a multilingual contrastive learning objective (Babu et al., 2022), whereas WavLM incorporates masked speech prediction together with denoising and speech enhancement objectives (Chen et al., 2022). These complementary learning paradigms motivate combining the two encoders through score-level fusion.

For each input waveform, the XLS-R-300M and WavLM-Large front-ends produce contextual frame-level representations through their Transformer encoder layers. Rather than using only the final-layer representation, we apply a learnable softmax-weighted aggregation across the encoder layers to capture information at different representation levels. The resulting 1024-dimensional SSL features are projected to 128 dimensions and passed to the AASIST back-end for spoofing detection.

The projected features are passed to AASIST (Jung et al., 2022), whose spectral and temporal graph-attention branches produce spoofing representations for binary bona fide/spoof classification. The utterance-level spoofing score is computed as the difference between the bona fide and spoof logits.

3.2 Domain Adaptation

Motivated by a pronounced telephone-band mismatch observed between the training and evaluation audio, we apply telephony channel augmentation during training by exposing the models to bandwidth-limited and codec-degraded speech. A detailed spectral analysis of this mismatch is provided in Appendix 7.2. We further employ iterative pseudo-label self-training. After each training round, both models independently score the unlabeled evaluation recordings, and only samples receiving consistent, high-confidence predictions are incorporated into the next training round as pseudo-labeled data. Each iteration is warm-started from the best checkpoint of the previous round and retained only if it improves validation EER.

3.3 Score-Level Fusion

The two SSL+AASIST models are trained independently, and their normalized prediction scores are combined using a weighted linear fusion. Compared with feature-level fusion, this approach preserves the independently learned representations without requiring joint end-to-end optimization.

For each utterance, prediction scores are first normalized using z-score normalization and subsequently fused using

$$S_{\text{fusion}} = wS_{\text{XLSR}} + (1 - w)S_{\text{WavLM}}, \qquad (1)$$

where w is selected by minimizing the EER on the validation set.

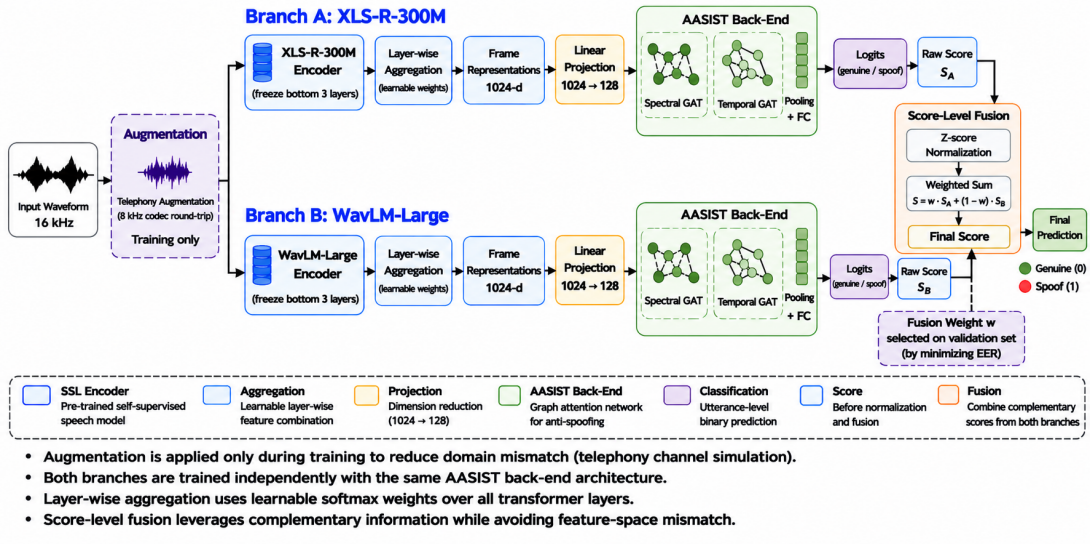


Figure 1: Proposed dual-branch SSL+AASIST architecture with weighted score-level fusion.

4 Experimental Setup

4.1 Preprocessing

All recordings are converted to mono and resampled to 16 kHz. For training, each recording is segmented into fixed-length 6-second speech segments, which are used as model inputs. During inference, recordings longer than 6 seconds are processed using consecutive 6-second windows, and the window-level prediction scores are averaged to obtain the final utterance-level score.

4.2 Training Configuration

Both SSL encoders are initialized from publicly available pretrained checkpoints, while the AASIST back-end is initialized from weights pretrained on the ASVspoof 2019 LA dataset (Jung et al., 2022; Todisco et al., 2019). Models are trained on NVIDIA H200 GPUs using AdamW with cosine annealing, mixed-precision training, inverse-frequency class weighting, and cross-entropy loss.

The SSL encoders and AASIST back-ends are optimized using separate learning rates. Telephony channel augmentation using 8 kHz G.711/GSM codec round-trips is applied during training. Pseudo-label self-training is performed iteratively, with each round warm-started from the previous checkpoint and retained only if it improves validation EER. Grid search on the validation set selected fusion weights of 0.53 and 0.47 for XLS-R and WavLM, respectively. Additional optimization and implementation details are provided in Appendix 7.1.

Table 1: Validation performance of the individual SSL models and the final score-level fusion on the reconstructed validation set (13,428 segments).

Model	Validation EER (%)
XLS-R-300M	2.43
WavLM-Large	3.50
Score-level Fusion	**2.24**

Table 2: Validation EER (%) across successive pseudo-label self-training rounds. Values correspond to the validation EER recorded during iterative training; final selected models were subsequently rescored on the full validation set.

Model	Init	R1	R2	R3	R4
XLS-R-300M	–	4.47	3.36	2.63	**2.44**
WavLM-Large	–	4.87	**3.47**	–	–

5 Results and Analysis

5.1 Performance Overview

Our final official submission achieved an EER of **0.9265%** on the ArA-DF 2026 Track 1 evaluation set, ranking **4th** on the official leaderboard. Table 1 compares the individual SSL models and their score-level fusion on the reconstructed validation set, while Table 2 summarizes the effect of successive pseudo-label self-training rounds. On the internal validation set, fusion reduces validation EER from 2.43% for XLS-R to 2.24%, indicating complementary information across the two SSL encoders.

Successive pseudo-label rounds reduced validation EER for both encoders, with XLS-R reaching 2.44% after four rounds and WavLM reaching

Table 3: Ablation study on the official Track 1 evaluation set comparing individual SSL+AASIST models with the proposed score-level fusion. ROC-AUC and EER are threshold-independent metrics, while accuracy is reported at the corresponding EER operating threshold.

Model	ROC-AUC	EER (%)	Accuracy@EER (%)
XLS-R	0.9995	0.88	99.12
WavLM	0.9973	2.03	97.97
Fused	0.9993	0.93	99.07

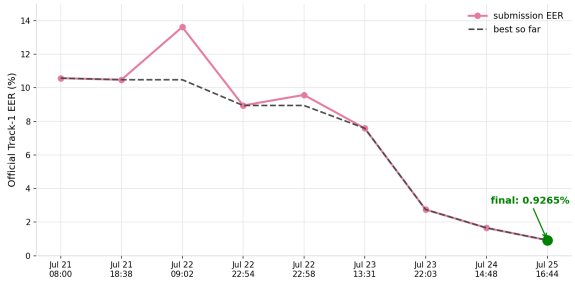


Figure 2: Official Track 1 EER across our nine submissions. Each point represents one leaderboard submission, and the dashed line indicates the best score obtained up to that point. The final selected submission achieved an EER of 0.9265%.

3.47% after two rounds. Further details on the iterative training procedure and stopping criterion are provided in Appendix 7.1.

5.2 System Comparison and Ablation

Table 3 compares the individual SSL models and their fusion. XLS-R provides the strongest standalone performance with 0.88% EER, compared with 2.03% for WavLM. Interestingly, although fusion improves the internal validation EER from 2.43% to 2.24%, its EER in this comparison (0.93%) is marginally higher than XLS-R alone. This suggests that validation-optimized fusion does not necessarily outperform its strongest member under distribution shift.

At the EER operating threshold of 0.4697, the fused system achieves 99.07% bona fide recall and 99.07% spoof recall. Of the 36,903 bona fide samples, 342 are misclassified as spoofed, while 993 of the 107,307 spoofed samples are misclassified as bona fide. This corresponds to an overall accuracy of 99.07% at the EER operating point. Since the official evaluation set does not provide dialect-level labels, our error analysis is limited to aggregate class behavior and acoustic/channel characteristics.

5.3 Development Trajectory

Figure 2 illustrates the evolution of Track 1 EER across our nine leaderboard submissions. Starting from an SSL+AASIST baseline with an EER above 10%, the system showed substantial performance improvements during development. The largest improvement coincided with the introduction of telephony channel augmentation, while subsequent pseudo-label self-training further improved performance. Since these submissions reflect iterative system development rather than controlled single-factor experiments, the trajectory should be interpreted as a development log rather than a formal ablation.

5.4 Validation–Evaluation Gap

Despite the distribution mismatch between the reconstructed validation split and the hidden evaluation set, improvements in validation EER generally translated to improved leaderboard performance, largely because telephony augmentation was also applied to the validation set to better approximate the target channel conditions. This supported the use of validation EER for checkpoint selection and fusion-weight tuning. Moreover, Table 1 shows that score-level fusion outperformed either individual SSL encoder on the internal validation set, supporting the use of complementary representations during model development.

6 Conclusion

We presented a domain-adaptive SSL+AASIST framework for Arabic audio deepfake detection using score-level fusion, telephony augmentation, and iterative pseudo-label self-training. Our final submission achieved an EER of **0.9265%**, ranking **4th** on the ArA-DF 2026 Track 1 leaderboard. Fine-grained dialect analysis was limited by the absence of dialect labels in the evaluation data; future work will investigate robustness across individual dialects and broader recording conditions.

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7 Appendix

7.1 Additional Training Details

Optimization configuration. The SSL encoders and AASIST back-ends are optimized using separate learning rates. XLS-R uses 3×10^{-6} and 3×10^{-5} for the encoder and back-end, respectively, while WavLM uses 4×10^{-6} and 4×10^{-5}. Both models are trained with batch size 32, weight decay 10^{-4}, and the bottom three transformer layers frozen during fine-tuning. Models are trained for 2–3 epochs, with checkpoint selection based on the best validation EER.

Learning-rate scheduling. The learning rate is scheduled using cosine annealing, with the minimum learning rate set to

$$\eta_{\min} = \frac{\eta_{\mathrm{SSL}}}{20}, \tag{2}$$

where η_{SSL} denotes the initial learning rate of the corresponding SSL encoder.

Telephony augmentation. Telephony augmentation is applied randomly to the training set to increase channel diversity. For the validation set, augmentation is applied deterministically using a path-hashed random number generator, ensuring reproducible validation EER while maintaining a channel condition representative of the target domain.

Pseudo-label self-training. Pseudo-label self-training is performed iteratively for each SSL+AASIST model. At each round, the current model is used to generate predictions for the evaluation recordings, and samples satisfying the high-confidence selection criterion are incorporated as pseudo-labeled training data for the subsequent round. Each round is warm-started from the best checkpoint of the preceding round, and validation EER is used to determine whether the updated model is retained. For XLS-R, validation EER continued to improve through four rounds, reaching 2.44%. A fifth round did not improve upon the fourth round and was therefore discarded. For WavLM, the best validation EER of 3.47% was obtained after two rounds.

Implementation. The implementation uses PyTorch 2.2, torchaudio 2.2, and Hugging Face Transformers 4.43. Audio preprocessing is performed using FFmpeg.

7.2 Spectral Analysis of Training and Evaluation Data

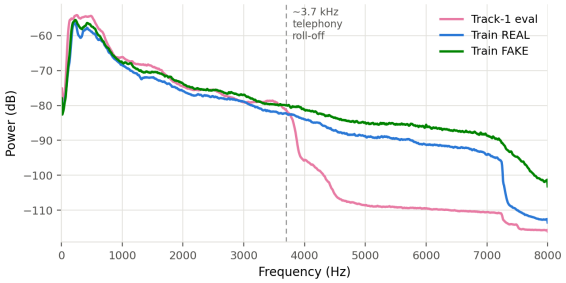


Figure 3: Average spectral characteristics of the reconstructed training data and the official evaluation recordings. The evaluation data exhibits a pronounced telephone-band spectrum with substantially reduced high-frequency energy, indicating a significant channel mismatch.

Figure 3 compares the average spectral characteristics of the reconstructed development data and the official evaluation recordings. The evaluation data exhibits a clear telephone-band response with substantial attenuation above approximately 4 kHz, revealing a pronounced channel mismatch. This observation motivated the use of telephony-aware augmentation during training.

To ensure that bandwidth itself could not serve as a shortcut for classification, the augmentation was applied equally to both bona fide and spoofed recordings, encouraging the detector to learn spoofing artifacts that remain robust across different

recording conditions.

Despite the pronounced channel mismatch, bandwidth itself is not discriminative of bona fide and spoofed speech. A simple classifier based only on high-frequency energy yields an EER of approximately 48%, close to chance. Thus, the benefit of telephony augmentation arises from reducing channel mismatch rather than providing a trivial bandwidth-based cue.